Risk II: Comparing Risks, Valuing the Contract

Advanced Microeconomics 1: Economic Decision Making. Lecture 8

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Where we left the question

- Monday built the machinery: expected utility, risk aversion as concave u, and the price of a risk — its certainty equivalent and premium.

- Two things remain. **(a)** We can price a *given* lottery, but not yet say when one distribution is *better* or *riskier* than another — needed, because the poor and the rich face *different* inflation lotteries.

- **(b)** The contract is still unvalued: indexation transforms the worker's lottery — less price risk, more job risk, from L6. Comparing transformed lotteries is exactly today's first tool.

Today's route

1. Comparing distributions: first- and second-order stochastic dominance.
2. The ex ante cost of inflation.
3. The insurance value of indexation.
4. When the assumptions are wrong — and what that means for our answers.
5. Conclusion.

Comparing distributions

Better: first-order stochastic dominance

Definition · FOSD

F **first-order stochastically dominates** G if for every nondecreasing $u : \mathbb{R} \to \mathbb{R}$, $\int u(x)\, dF(x) \geq \int u(x)\, dG(x)$.

- Equivalent test: $F(x) \leq G(x)$ for all x — F puts more weight on high realizations, everywhere. And FOSD $\Rightarrow \int x\, dF \geq \int x\, dG$.

- Read it as: *anyone who prefers more money to less* prefers F, whatever their risk attitude. A strong result!

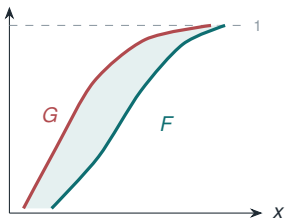
Riskier: second-order stochastic dominance

Definition · SOSD

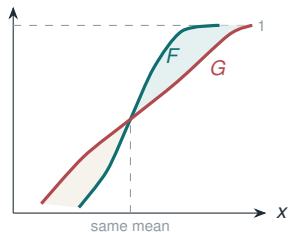
For F, G with the *same mean*: F **second-order stochastically dominates** G if for every nondecreasing *concave* u, $\int u(x)\,dF(x) \geq \int u(x)\,dG(x)$.

- G is a **mean-preserving spread** of F. Same mean, more dispersion.
- This is the formal content of ”riskier”: *every* risk-averse agent ranks F above G. Where FOSD ordered *levels*, SOSD orders *risk* (variance) at fixed level.

In pictures



FOSD: $F(x) \leq G(x)$ *everywhere* — no crossing. F's mass sits to the right: anyone who likes money prefers F.



SOSD: one crossing, *equal shaded areas* $\Leftrightarrow$ same mean. G moves mass to both tails — a mean-preserving spread: every risk-averter prefers F.

The ex ante cost of inflation

The setup: 2022 as a lottery, seen from December 2021

- Two states, prior $\Pr(\text{shock}) = 0.2$ (illustrative):

	normal ($p = 0.8$)	shock ($p = 0.2$)
prices	uniform $\pi = 2\%$	the 2022 draw ($g_E = 57.9$, $\pi_O = 6.3$)
bill, poor / rich	2.0% / 2.0%	11.3% / 9.3% (L6's full-price bills)

- NB the normal column: a uniform 2% costs every basket exactly 2%.

- Preferences: CRRA, with $r \in [1, 3]$. Each household's real income is a lottery; its **risk premium** is the ex ante cost of inflation risk.

What would you pay ex-ante to not face the risk of 2022 inflation

risk premium of the uninsured lottery	$r = 1$	$r = 2$	$r = 3$
poor ($s_0 = 0.10$)	0.07%	0.15%	0.23%
rich ($s_0 = 0.06$)	0.05%	0.09%	0.14%

- **Ordering**: the poor's premium exceeds the rich's at every r — their lottery is the rich's, shifted down *and* spread out (bigger exposure to the relative-price draw).

- **Magnitude**: 0.05–0.25% of income — an order below the realized bill ($\sim$ 10%). What hurt was overwhelmingly the draw, not the uncertainty.

Proof → p. 25

The insurance value of indexation

The two lotteries a worker can hold

Previous calculations:

real income, by state	uninsured	indexed
normal ($p = 0.8$)	-2%	0
shock ($p = 0.2$)	-11.3 / -9.3%	-1.7 / $+0.3\%$

Lotteries available:

		mean	variance ($\%^2$)
poor	uninsured	-3.86%	13.84
	indexed	-0.34%	0.46
rich	uninsured	-3.46%	8.53
	indexed	$+0.06\%$	0.01

- NB, at the moment ignore the job loss channel ($+4.5$pp under indexation).

The insurance value of indexation

insurance value ΔCE ($r = 2$)	total	mean part	risk part
poor	$+3.67\%$	$+3.52\%$	$+0.15\%$
rich	$+3.61\%$	$+3.52\%$	$+0.09\%$

- **The contract is valuable** — worth $\approx +3.6\%$ of income ex ante.
- Almost all of the value is the mean part: a state-contingent transfer that pays when times are bad. The pure risk-removal part is 0.1–0.15%.
- Unpriced: Unemployment risk.

Proof → p. 26

When the assumptions are wrong

Failure I: independence (Allais) — what it does to our numbers

- Why it matters here: our results are certainty-equivalent numbers, and CE calculations require the linearity that Allais invalidates. If certainty is overweighted, the value of guaranteed indexation is higher than expected utility says — and fear of the uninsured tail is stronger.

- Allais-type behaviour raises the felt value of automatic indexation but would not overturn the decomposition.

Failure II: state-independent utility

- Utility is plausibly **state-dependent**: $u(x; \text{employed}) \neq u(x; \text{workless})$ — job loss changes health, time, status, not just the budget.

- If marginal utility is higher when workless, then: (i) the job-hazard cost of indexation is larger; (ii) symmetrically, transfers that pay in bad states are worth more — insurance equalises marginal utility across states.

- Direction: state-dependence amplifies both sides — the insurance value and the jobs price — and shifts the design question toward: insure the workless directly (benefits, already indexed) rather than only the employed's wage.

Failure III: objective probabilities

- We assumed the prior is known. Two failures.
- **Ellsberg**: offered bets on a known 50/50 urn vs an urn of unknown composition, most prefer the known — no single subjective probability rationalises the pattern. Ambiguity aversion: people dislike not knowing the odds.
- **Measured inflation beliefs**: household inflation expectations are systematically biased upward and enormously dispersed, anchored on salient prices like groceries and fuel (survey evidence; Weber et al. 2022).
- Direction: if households cannot assess π-risk, they will not buy the right insurance privately. Ambiguity aversion $+$ ambiguity over inflation risk $\Rightarrow$ insurance is worth more.

To index or not to index, that is the question

Should we index?

margin	block	tool	the number
transfer accuracy, ex post	1	CV, Stone–Geary	bills 11.3/9.3 vs cheque 9.6 $\Rightarrow -1.7/+0.3$
incidence, through firms	2	Shephard, CES	$+1.1$pp prices, -4.5% jobs,
insurance, ex ante	3	EU, CRRA	risk 0.05–0.25%; mean $+3.3\%$

Roughly:

- Indexation undoes the € cost of inflation despite causing it to be a little worse.
- Average protection hides regressive heterogeneity.
- Cost of this protection is unemployment.
- Hidden benefit as an insurance is non-negligible but small.

What we missed

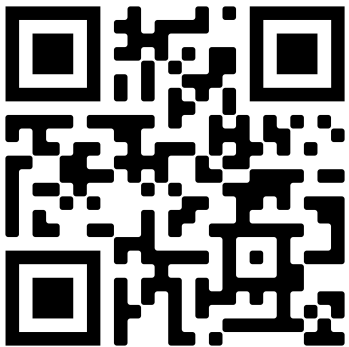
- The GE effects of higher real wages pushing up demand. Pro-indexation.
- Input-Ouput feedback of intermediate goods carring the wage rise. Anti-indexation.
- Open-economy. Labor gets more expensive, domestic goods get more expensive, switch to imports. Anti-indexation.
- Heterogeneity is on many other dimensions than those we study.

What we assumed

- We made many assumptions to come to our answer.
- These will not necessarily be the right assumptions...
- They should be scrutinised, and the sensitivity of our conclusion to them quantified.
- Only because we were rigorous and formal can we clearly articulate our assumptions, and their empirical content, and therefore stress-test them.

Fin

Feedback :)



Appendix

Derivations

The uninsured poor household's lottery over gross real income x: $x = 0.98$ w.p. 0.8, $x = 0.887$ w.p. 0.2. Four steps, at $r = 2$ (so $u(x) = -1/x$, $u^{-1}(v) = -1/v$):

1. **Mean**: $\mathbb{E}[x] = 0.8(0.98) + 0.2(0.887) = 0.9614$.
2. **Expected utility**: $\mathbb{E}[u] = -\left(\frac{0.8}{0.98} + \frac{0.2}{0.887}\right) = -(0.81633 + 0.22548) = -1.04181$.
3. **Invert**: $c = -1/\mathbb{E}[u] = 1/1.04181 = 0.95987$.
4. **Premium**: $\mathbb{E}[x] - c = 0.9614 - 0.95987 = 0.00153 \Rightarrow$ **0.15**% of expenditure.

Other cells, same recipe: $r = 1$ uses ln: $\mathbb{E}[\ln x] = -0.04014$, $c = e^{-0.04014} = 0.96065 \Rightarrow 0.07\%$. $r = 3$ uses x^{-2}: $\mathbb{E}[x^{-2}] = 1.08719$, $c = 1.08719^{-1/2} = 0.95907 \Rightarrow 0.23\%$. Rich at $r = 2$: outcomes $0.98/0.907$, $c = 0.96448 \Rightarrow 0.09\%$.

← Back to p. 10

Derivations

Run the same four steps on the *indexed* lottery (poor: $x = 1.0$ w.p. 0.8, $x = 0.983$ w.p. 0.2), at $r = 2$:

$$\mathbb{E}[x] = 0.9966, \qquad \mathbb{E}[u] = -\left(0.8 + \tfrac{0.2}{0.983}\right) = -1.00346, \qquad c = 0.99655, \qquad RP = 0.00005$$

1. **Total**: $\Delta CE = c_{\text{idx}} - c_{\text{unins}} = 0.99655 - 0.95987 = 0.03668 \;\Rightarrow\; +\mathbf{3.67}\%$.
2. **Decompose** via $CE = \text{mean} - RP$, so

$$\Delta CE = \underbrace{\Delta\text{mean}}_{0.9966-0.9614\,=\,+3.52\text{pp}} + \underbrace{\left(RP_{\text{unins}} - RP_{\text{idx}}\right)}_{0.15\%-0.005\%\,=\,+0.15\text{pp}}$$

   the "risk part" *is* the difference of the two premia: what variance-removal is worth.
3. **Rich**, same steps: $c_{\text{idx}} = 1.00060$, $c_{\text{unins}} = 0.96448$, $\Delta CE = +3.61\%$; mean part identical ($+3.52$: the cheque is common), risk part $+0.09$.

← Back to p. 13